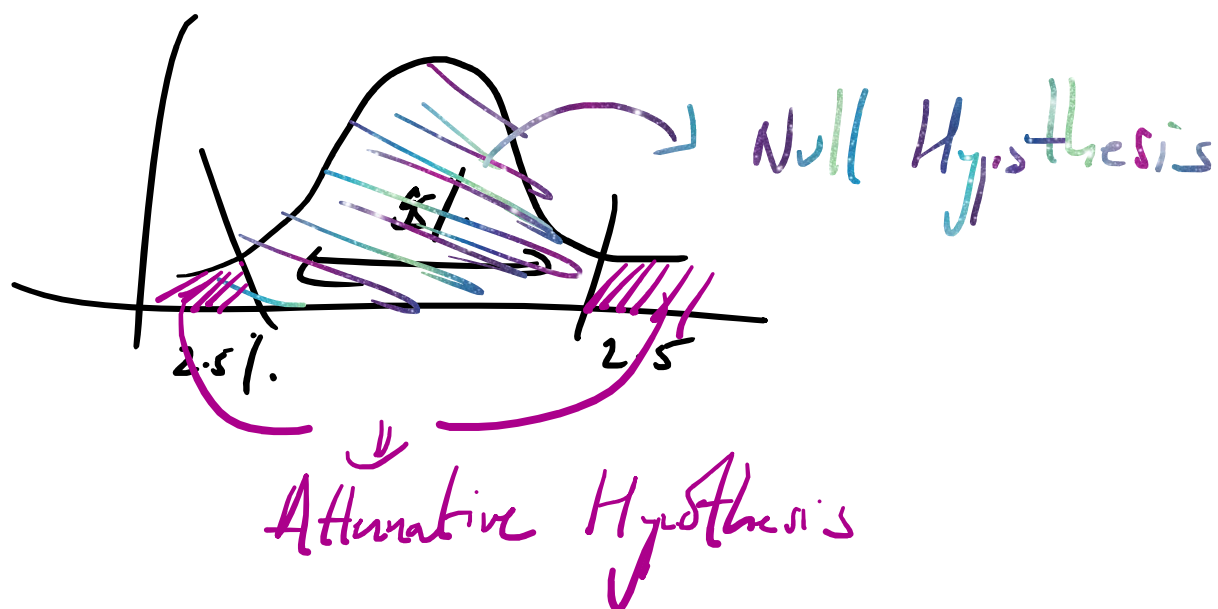


$\Rightarrow$ Hypothesis Testing : (Inference)

Null Hypothesis : H_0

Alternative Hypothesis : H_1 , H_a .

level of significance : Denote by α is probability of Type I error.



$\Rightarrow$ Mean Age of people in India is 30 years.

- $\rightarrow H_0: \mu = 30$ Against $H_1: \mu \neq 30$ (2-tail)
 $\rightarrow H_0: \mu = 30$ Against $H_1: \mu < 30$ (Left Tail)
 $\rightarrow H_0: \mu = 30$ Against $H_1: \mu > 30$ (Right Tail)

$\Rightarrow$ Critical Region: Uses level of significance
 for determining acceptance or rejection
 of Null Hypothesis.

$\Rightarrow$ Hypothesis Testing for Mean:

$\hookrightarrow$ Case I: σ is known.

Population is normally distributed or
 sample is large ($n \geq 30$).

$$\text{Test Statistic} = Z_c = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}$$

$\Rightarrow$ A survey done 10 years ago of CA income

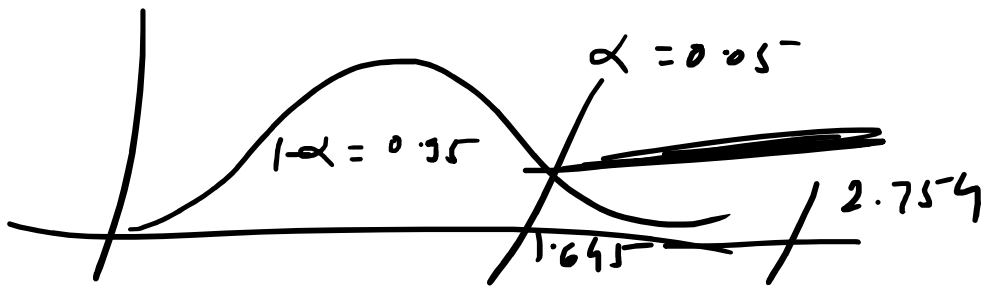
of a month is 74914 rs. A researcher wants to test whether over the years this average is increased or not. A sample of 112 CA produce mean salary of 78695/- . Also S.D $\sigma = \underline{\underline{14530/-}}$.

① $H_0 : \mu = 74914$ against $H_a : \mu > 74914$.
 $\therefore$ Right Tail Test will be executed.

② $n = 112$, $\sigma = 14530$, $\bar{x} = 78695$, $\mu = 74914$

$$\begin{aligned} \textcircled{3} \quad Z_c &= \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}} = \frac{78695 - 74914}{14530 / \sqrt{112}} \\ &= \underline{\underline{2.754}} \end{aligned}$$

④



cum. prob	$t_{.50}$	$t_{.75}$	$t_{.80}$	$t_{.85}$	$t_{.90}$	$t_{.95}$	$t_{.975}$	$t_{.99}$	$t_{.995}$	$t_{.999}$	$t_{.9995}$
→ one-tail	0.50	0.25	0.20	0.15	0.10	0.05	0.025	0.01	0.005	0.001	0.0005
two-tails	1.00	0.50	0.40	0.30	0.20	0.10	0.05	0.02	0.01	0.002	0.001
df											
1	0.000	1.000	1.376	1.963	3.078	6.314	12.71	31.82	63.66	318.31	636.62
2	0.000	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.327	31.599
3	0.000	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	10.215	12.924
4	0.000	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	0.000	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	0.000	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	0.000	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	0.000	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	0.000	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	0.000	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	0.000	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	0.000	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	0.000	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	0.000	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	0.000	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	0.000	0.690	0.865	1.071	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	0.000	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	0.000	0.688	0.862	1.067	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	0.000	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	0.000	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.552	3.850
21	0.000	0.686	0.859	1.063	1.323	1.721	2.080	2.518	2.831	3.527	3.819
22	0.000	0.686	0.858	1.061	1.321	1.717	2.074	2.508	2.819	3.505	3.792
23	0.000	0.685	0.858	1.060	1.319	1.714	2.069	2.500	2.807	3.485	3.768
24	0.000	0.685	0.857	1.059	1.318	1.711	2.064	2.492	2.797	3.467	3.745
25	0.000	0.684	0.856	1.058	1.316	1.708	2.060	2.485	2.787	3.450	3.725
26	0.000	0.684	0.856	1.058	1.315	1.706	2.056	2.479	2.779	3.435	3.707
27	0.000	0.684	0.855	1.057	1.314	1.703	2.052	2.473	2.771	3.421	3.690
28	0.000	0.683	0.855	1.056	1.313	1.701	2.048	2.467	2.763	3.408	3.674
29	0.000	0.683	0.854	1.055	1.311	1.699	2.045	2.462	2.756	3.396	3.659
30	0.000	0.683	0.854	1.055	1.310	1.697	2.042	2.457	2.750	3.385	3.646
40	0.000	0.681	0.851	1.050	1.303	1.684	2.021	2.423	2.704	3.307	3.551
60	0.000	0.679	0.848	1.045	1.296	1.671	2.000	2.390	2.660	3.232	3.460
80	0.000	0.678	0.846	1.043	1.292	1.664	1.990	2.374	2.639	3.195	3.416
100	0.000	0.677	0.845	1.042	1.290	1.660	1.984	2.364	2.626	3.174	3.390
1000	0.000	0.675	0.842	1.037	1.282	1.646	1.962	2.330	2.581	3.098	3.300
Z	0.000	0.674	0.842	1.036	1.282	1.645	1.960	2.326	2.576	3.090	3.291
	0%	50%	60%	70%	80%	90%	95%	98%	99%	99.8%	99.9%
	Confidence Level										

If $\alpha < z_c$; Reject Null Hypothesis

Accept Alternative Hypothesis

Accept Alternative Hypothesis

If $\alpha > z_c$; Reject Alternative Hypo

↓
Accept Null Hypothesis.

$$\alpha < z_c$$

$$1.645 < 2. \dots$$

⑤ Verdict: Since test statistic has proven to accept the alternative hypothesis it claims that the salary over the years has increased.

⇒ Case 2: σ is not known:

$$T_c = \frac{\bar{x} - \mu_0}{s_1 / \sqrt{n}}$$

$$s_1 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

$$s_1 = \frac{1}{n-1} \sum_{i=1}^n \sqrt{(x_i - \bar{x})^2}$$

$\Rightarrow$ A sample of 25 people is taken. The length of time to prepare dinner is recorded in minutes.

44, 51.9, 49.7, 40, 55.5, 33, 43.4, 41.3,
45.2, 40.7, 41.1, 45.1, 30.5, 45.2, 55.3, 52.1,
55.1, 38.8, 43.1, 35.2, 58.6, 49.8, 43.2,
47.9, 46.6.

Is there evidence that mean time to prepare dinner is less than 48 mins?

$\Rightarrow$ ① $H_0: \mu \geq 48$ against $H_a: \mu < 48$.

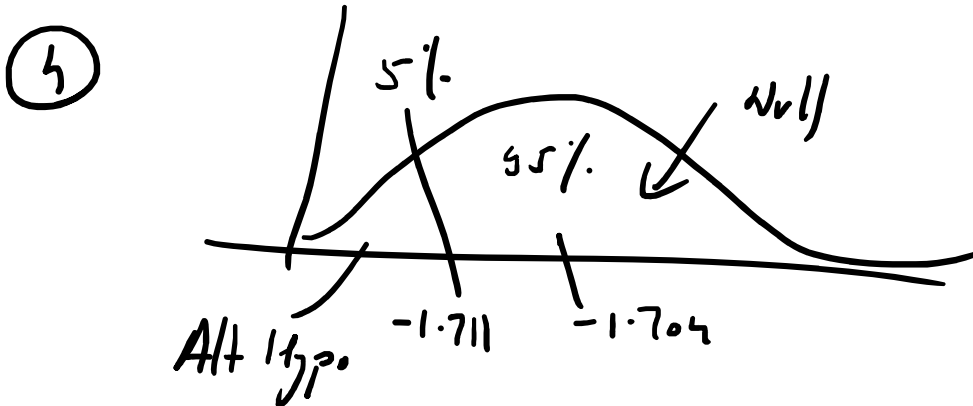
This proves the left-tailed test.

② $n = 25$, $\bar{x} = 45.628$, $s_1 = 6.9587$

③ Test Statistic

③ Test Statistic

$$T_c = \frac{\bar{x} - \mu_0}{s_1 / \sqrt{n}} = \frac{45.628 - 48}{6.5587 / \sqrt{25}} = -1.704$$



cum. prob	t _{.50}	t _{.75}	t _{.80}	t _{.85}	t _{.90}	t _{.95}	t _{.975}	t _{.99}	t _{.995}	t _{.999}	t _{.9995}
one-tail	0.50	0.25	0.20	0.15	0.10	0.05	0.025	0.01	0.005	0.001	0.0005
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	0%	50%	60%	70%	80%	90%	95%	98%	99%	99.8%	99.9%
Confidence Level											

$$\alpha > T_c$$

$$|-1.711| > |-1.704|$$

$$1.711 > 1.704$$

$\therefore$ Reject the alternative Hypo & Accept the Null Hypo.

⑤ Verdict: Based on the hypothesis it has been claimed that the mean dinner preparation time is either 48 or more.